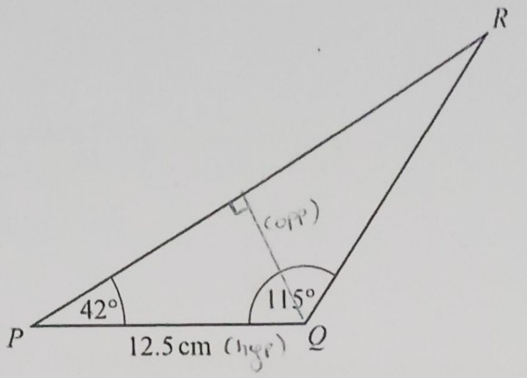
12



NOT TO SCALE

The diagram shows triangle PQR.

Calculate the shortest distance from Q to PR.

$\sin\theta = \frac{opp}{hyp}$

$\sin 42° = \frac{opp}{12.5}$

$\sin 42 \times 12.5 = opp$

$8.36\text{cm} = opp$

.........8.36......... cm [3]

13 Make x the subject of this formula.

$$A = w^2 + 5x^2$$

$\frac{A - w^2}{5} = \frac{5}{5}x^2$

$\frac{A - w^2}{5} = x^2$

$\pm\sqrt{\frac{A - w^2}{5}} = x$

$x =$ $\pm\sqrt{\frac{A - w^2}{5}}$ .... [3]